№2

: SIR -

2026-09-04

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1.
2. SIR
3. -
4.
5.
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1

- SIR ( )

- SIR ( )
- - (-)

- SIR ( )
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- Julia (DifferentialEquations.jl)

- SIR ( )
- - (-)
- Julia (DifferentialEquations.jl)
- (Literate.jl)

- SIR ( )
- - (-)
- Julia (DifferentialEquations.jl)
- (Literate.jl)
- Jupyter Notebook Quarto-

2

SIR

SIR —

:

$$\frac{dS}{dt} = -\beta c \frac{IS}{N}, \quad \frac{dI}{dt} = \beta c \frac{IS}{N} - \gamma I, \quad \frac{dR}{dt} = \gamma I$$

β	(0–1)
c	
γ	
$R_0 = c\beta/\gamma$	

SIR

β	0.05
c	10
γ	0.25
R_0	**2.0**
S_0, I_0, R_0	990, 10, 0

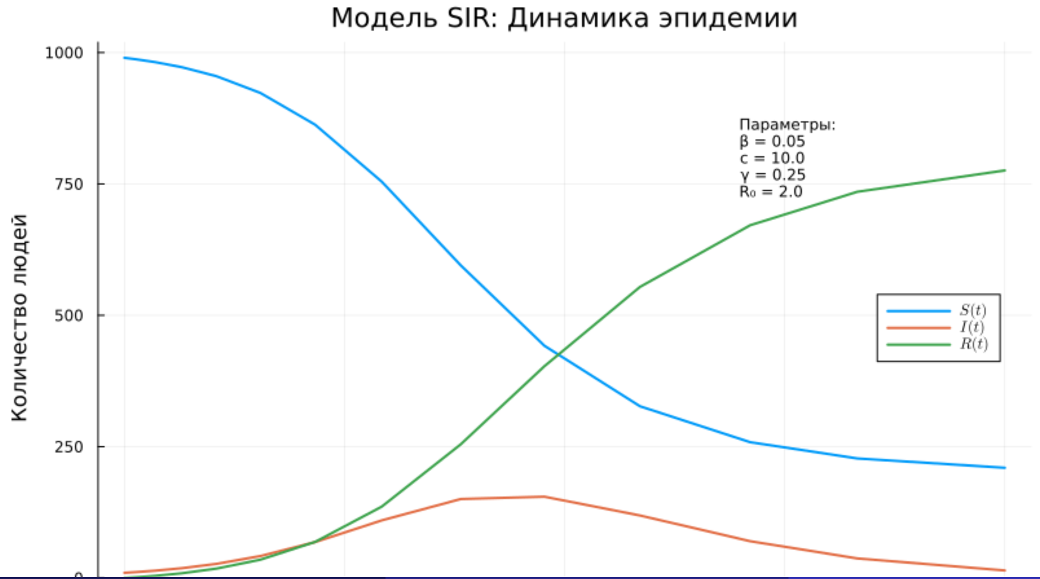
SIR

```
lilidji@eiglushchenko:~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project$ julia --project=. scripts/sir_ode.jl
Параметры модели SIR:
β (вероятность заражения) = 0.05
c (среднее число контактов) = 10.0
γ (скорость выздоровления) = 0.25
R₀ = c * β / γ = 2.0
Средняя продолжительность болезни = 4.0 дней
Начальные условия: S₀ = 990.0, I₀ = 10.0, R₀ = 0.0
Could not create decoration from factory! Running with no decorations.

Бенчмарк решения:


=== АНАЛИЗ РЕЗУЛЬТАТОВ ===
Общая численность популяции (контроль): N = 1000.0
Пиковое число зараженных: I_max = 154.8
Время достижения пика: t_peak = 19.1 дней
Итоговое число переболевших: R(∞) = 775.7
Доля переболевших: 77.6%

Теоретический анализ:
  - Порог коллективного иммунитета: 50.0%
  - Теоретический пик при S/N = 1/R₀ = 0.5
```

1: : julia –project=. scripts/sir_ode.jl

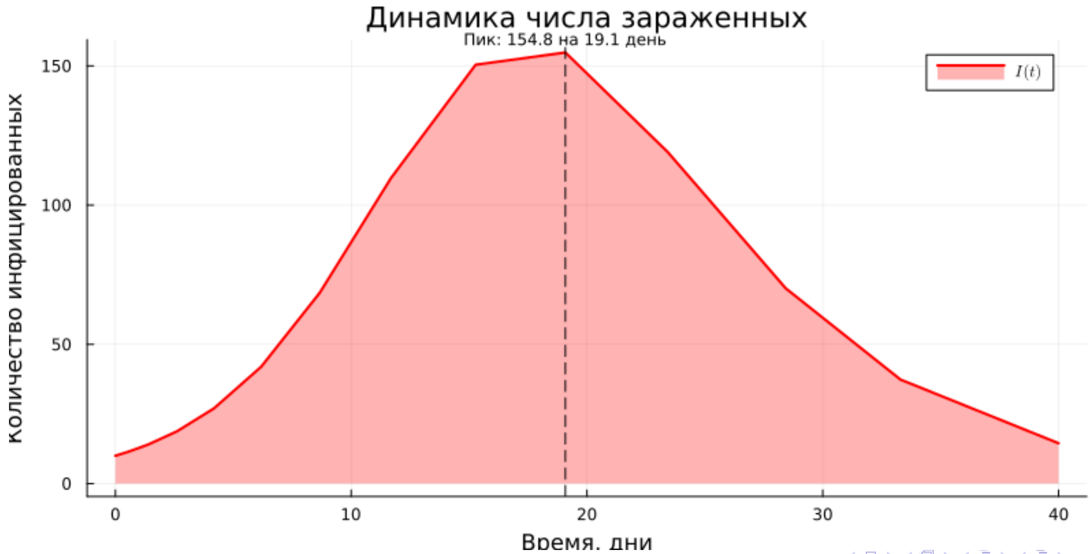
SIR

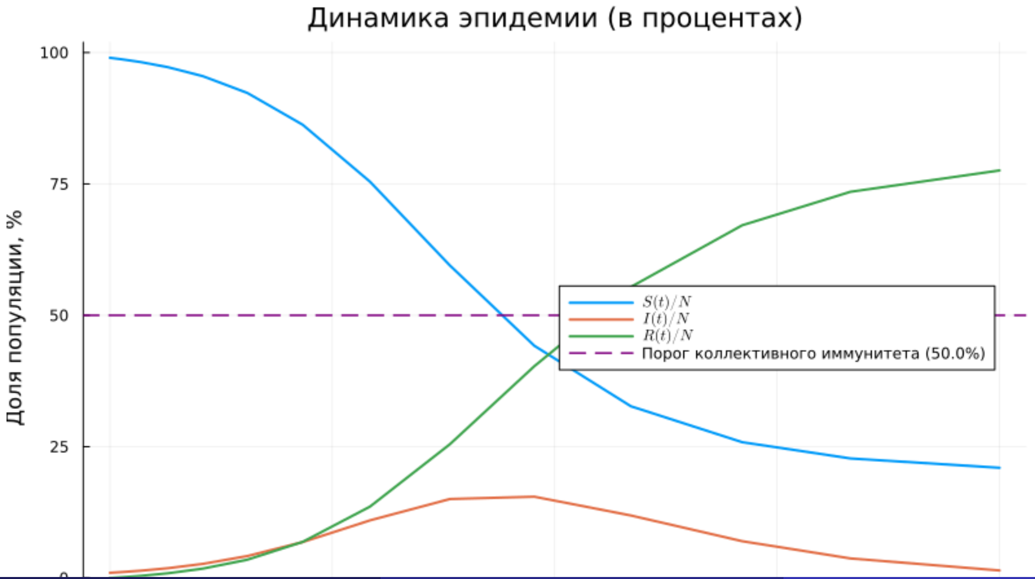
I_{max}	154.8
	19.1
$R(\infty)$	775.7 (77.6%)
	50%



Модель SIR: Динамика эпидемии
Количество людей
1000
750
500
250
0
Параметры:
β = 0.05
c = 10.0
γ = 0.25
R₀ = 2.0
S(t)
I(t)
R(t)

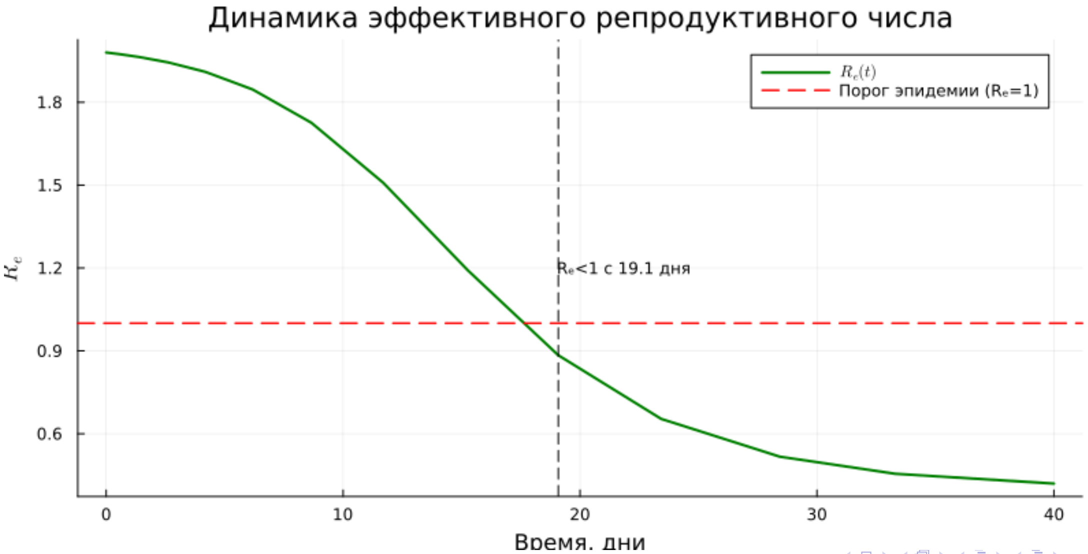
I(t)


Динамика числа зараженных
Пик: 154.8 на 19.1 день
150
100
50
0
0
10
20
30
40
Количество инфицированных
Время, дни
$I(t)$

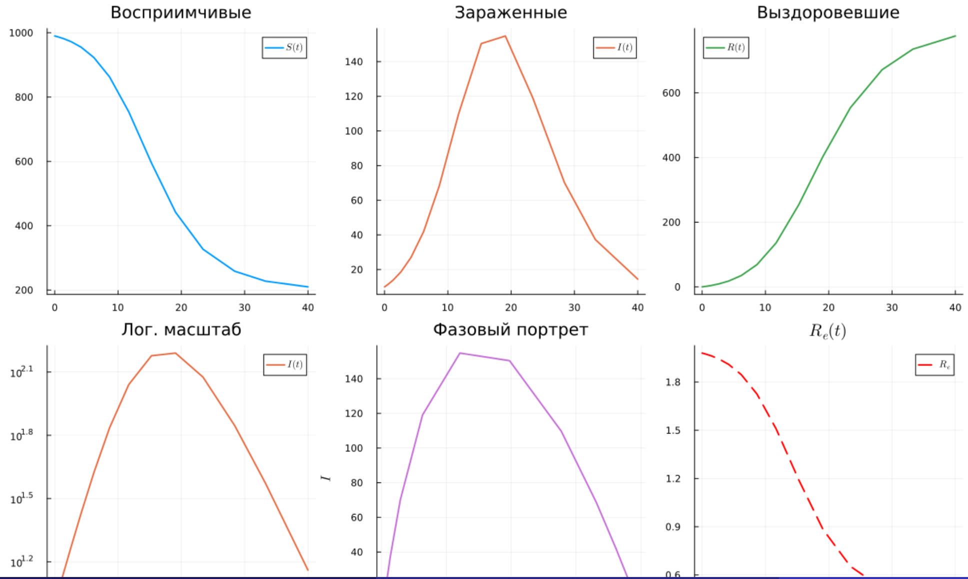

Динамика эпидемии (в процентах)
100
75
50
25
0
Доля популяции, %
$S(t)/N$
$I(t)/N$
$R(t)/N$
Порог коллективного иммунитета (50.0%)

SIR


Фазовый портрет SIR модели
Фазовая траектория
I(t)
140
120
100
80
60
40
20


Динамика эффективного репродуктивного числа
$R_e(t)$
Порог эпидемии (Rₑ=1)
Rₑ<1 с 19.1 дня
R_e
1.8
1.5
1.2
0.9
0.6
0
10
20
30
40
Время, дни

SIR



Восприимчивые
S(t)
1000
800
600
400
200
0
10
20
30
40
Зараженные
I(t)
140
120
100
80
60
40
20
Выздоровевшие
R(t)
600
400
200
Лог. масштаб
$10^{2.1}$
$10^{1.8}$
$10^{1.5}$
$10^{1.2}$
Фазовый портрет
I
$R_e(t)$
R_e
1.8
1.5
1.2
0.9
0.6

3

-

- —

- :

$$\frac{dx}{dt} = \alpha x - \beta xy, \quad \frac{dy}{dt} = \delta xy - \gamma y$$

α
β
δ
γ

: $x^* = \gamma/\delta$, $y^* = \alpha/\beta$

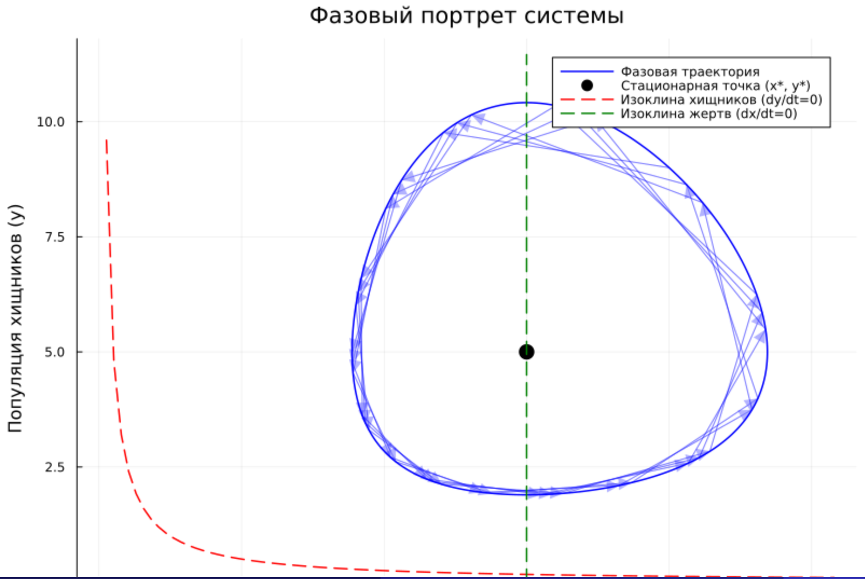
α	0.1
β	0.02
δ	0.01
γ	0.3
x_0, y_0	40, 9
x^*, y^*	30.0, 5.0

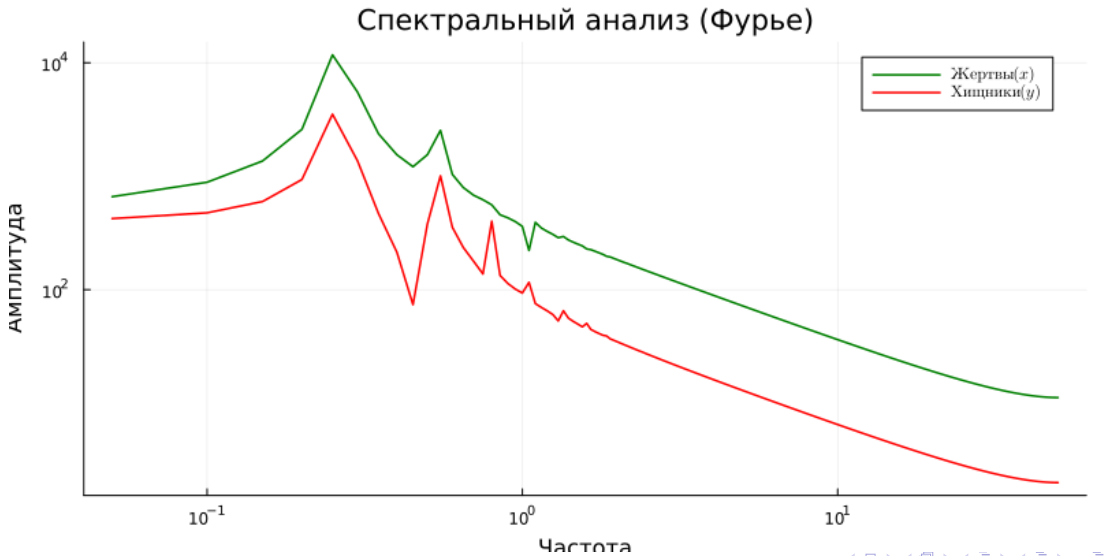
```
lilidji@eiglushchenko:~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project$ julia --project=. scripts/lv_ode.jl
============================================================
Модель Лотки-Вольтерры (хищник-жертва)
============================================================

Параметры модели:
α (скорость размножения жертв) = 0.1
β (скорость поедания жертв) = 0.02
δ (коэффициент конверсии) = 0.01
γ (смертность хищников) = 0.3

Начальные условия:
Жертвы (x₀) = 40.0
Хищники (y₀) = 9.0

Стационарные точки (положения равновесия):
x* = γ/δ = 30.0
y* = α/β = 5.0

Доминирующая частота колебаний жертв: 0.2499 Гц
Период колебаний жертв: 4.0 единиц времени

============================================================
Анализ результатов
============================================================

Основные статистики:
Жертвы: min = 17.75, max = 46.89, mean = 29.71
Хищники: min = 1.9, max = 10.41, mean = 5.2

Анализ колебаний:
Первый пик жертв: время = 33.7, значение = 46.89
Первый пик хищников: время = 2.9, значение = 10.41
Сдвиг фаз (хищники отстают): -30.8
Could not create decoration from factory! Running with no decorations.
```

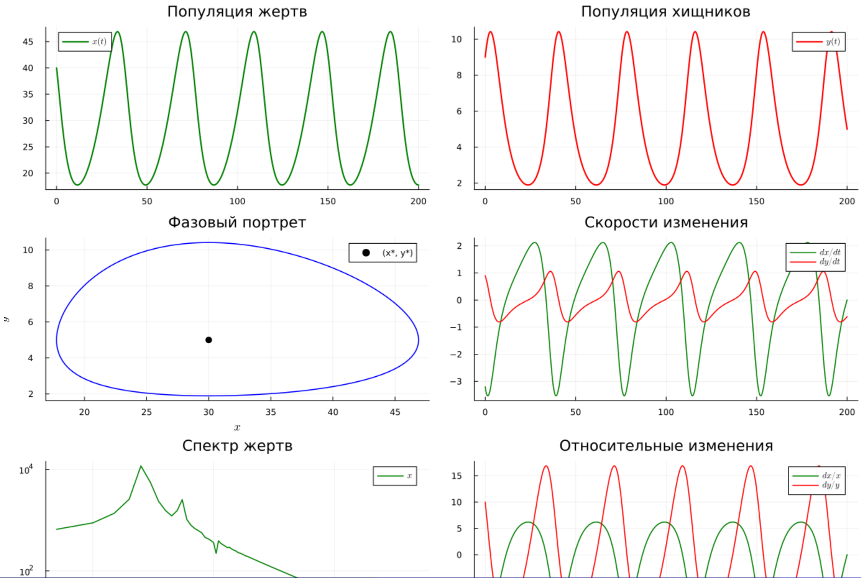
2: : julia –project=. scripts/lv_ode.jl

: / /	17.75 / 46.89 / 29.71
: / /	1.90 / 10.41 / 5.20
	~36.4 .


Модель Лотки-Вольтерры: Динамика популяций
Популяция
Жертвы(x)
Хищники(y)
x* (равновесие жертв)
y* (равновесие хищников)
10
20
30
40
0
50
100
150
200


Фазовый портрет системы
Популяция хищников (y)
10.0
7.5
5.0
2.5
Фазовая траектория
Стационарная точка (x*, y*)
Изоклина хищников (dy/dt=0)
Изоклина жертв (dx/dt=0)


Спектральный анализ (Фурье)
Жертвы(x)
Хищники(y)
10^4
10^2
Амплитуда
10^{-1}
10^0
10^1
Частота


Популяция жертв
$x(t)$
Популяция хищников
$y(t)$
Фазовый портрет
(x*, y*)
x
y
Скорости изменения
dx/dt
dy/dt
Спектр жертв
x
Относительные изменения
dx/x
dy/y

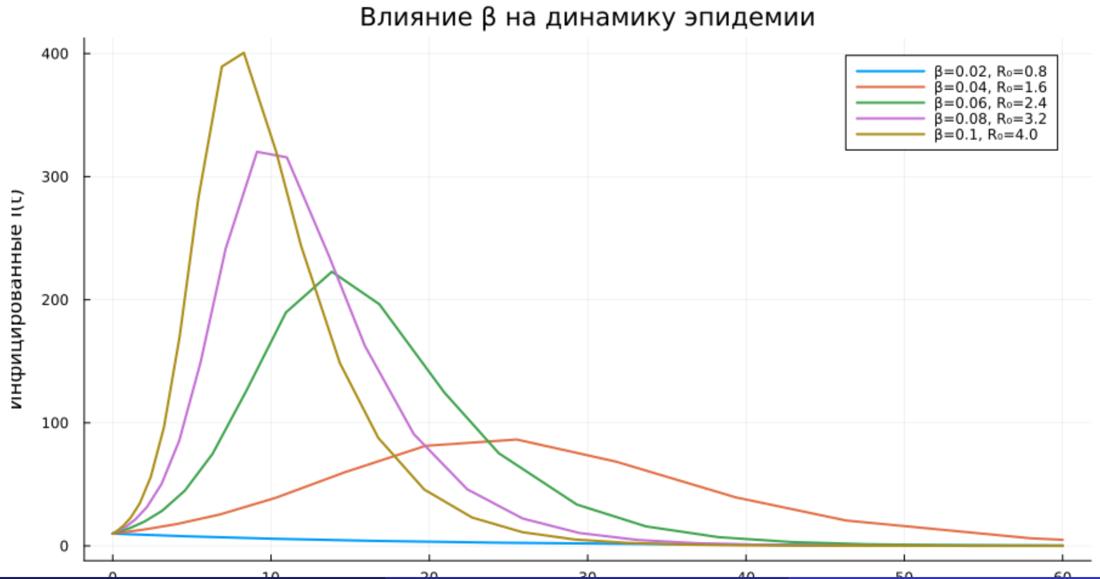
4

SIR

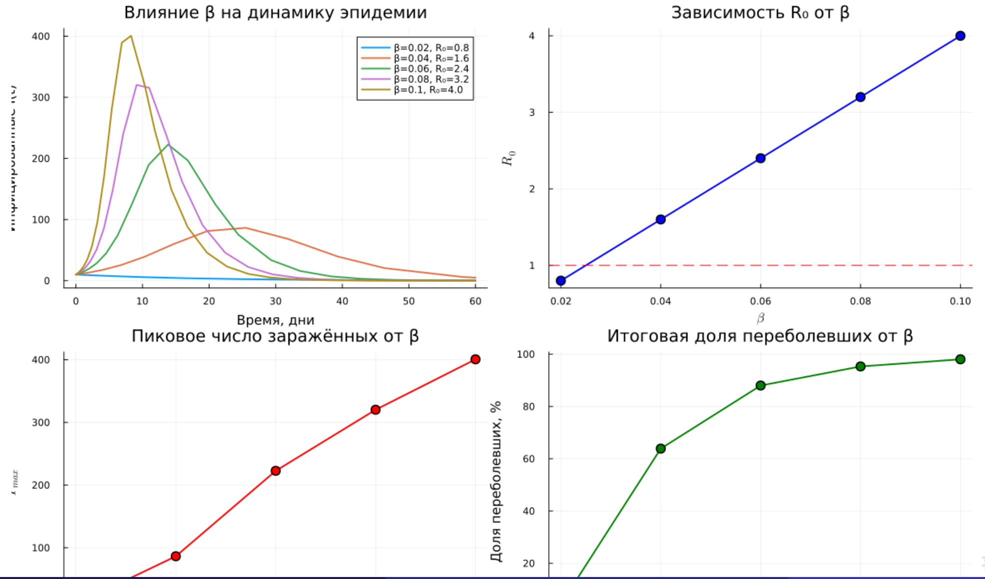
: $\beta \in [0.02, 0.04, 0.06, 0.08, 0.1]$

β	R_0	I_{max}	t_{peak}	$R(\infty)$, %
0.02	0.8	10.0	0.0	4.4
0.04	1.6	86.4	25.5	63.9
0.06	2.4	222.7	13.8	88.0
0.10	4.0	400.6	8.3	98.0

```
lilidji@eiglushchenko:~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project$ julia --project=. scripts/sir_ode_param.jl
============================================================
Параметрическое исследование модели SIR
============================================================
Фиксированные параметры: c = 10.0, γ = 0.25
Варьируемый параметр: β ∈ [0.02, 0.04, 0.06, 0.08, 0.1]

β = 0.02: R₀ = 0.8, I_max = 10.0, t_peak = 0.0 дн., R(∞) = 4.4%
β = 0.04: R₀ = 1.6, I_max = 86.4, t_peak = 25.5 дн., R(∞) = 63.9%
β = 0.06: R₀ = 2.4, I_max = 222.7, t_peak = 13.8 дн., R(∞) = 88.0%
β = 0.08: R₀ = 3.2, I_max = 320.2, t_peak = 9.1 дн., R(∞) = 95.3%
β = 0.1: R₀ = 4.0, I_max = 400.6, t_peak = 8.3 дн., R(∞) = 98.0%
Could not create decoration from factory! Running with no decorations.

============================================================
Сводная таблица результатов
============================================================
5×6 DataFrame
 Row │ β        R0       I_max     t_peak    R_final  R_final_pct
     │ Float64  Float64  Float64   Float64   Float64  Float64
─────┼──────────────────────────────────────────────────────────────
   1 │    0.02      0.8   10.0       0.0      43.641       4.3641
   2 │    0.04      1.6   86.4294  25.5143   638.717      63.8717
   3 │    0.06      2.4  222.742   13.8428   880.103      88.0103
   4 │    0.08      3.2  320.188    9.11419  953.108      95.3108
   5 │    0.1       4.0  400.627    8.27961  980.396      98.0396

Параметрическое исследование SIR завершено!
lilidji@eiglushchenko:~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project$ julia --project=. scripts/lv_ode_param.jl
```


Влияние β на динамику эпидемии
инфицированные I(t)
β=0.02, R₀=0.8
β=0.04, R₀=1.6
β=0.06, R₀=2.4
β=0.08, R₀=3.2
β=0.1, R₀=4.0
0
100
200
300
400
0
10
20
30
40
50
60

SIR

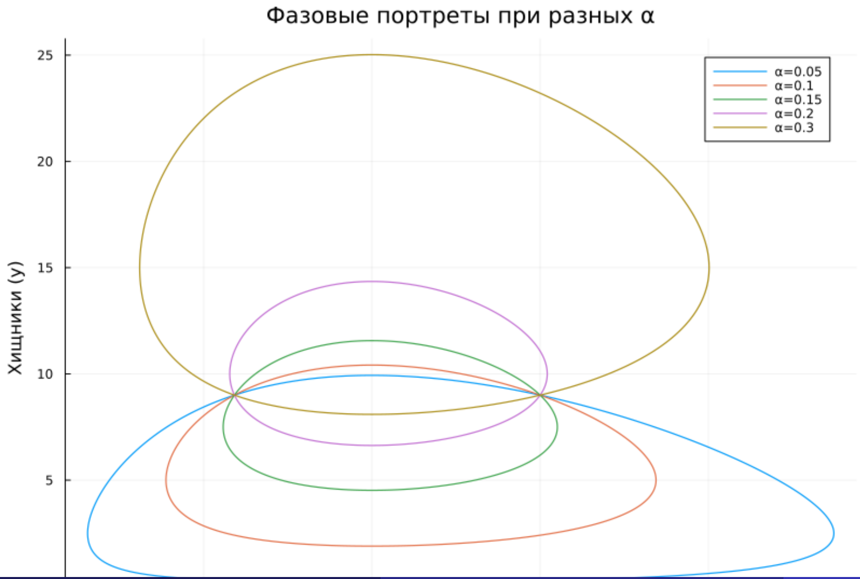


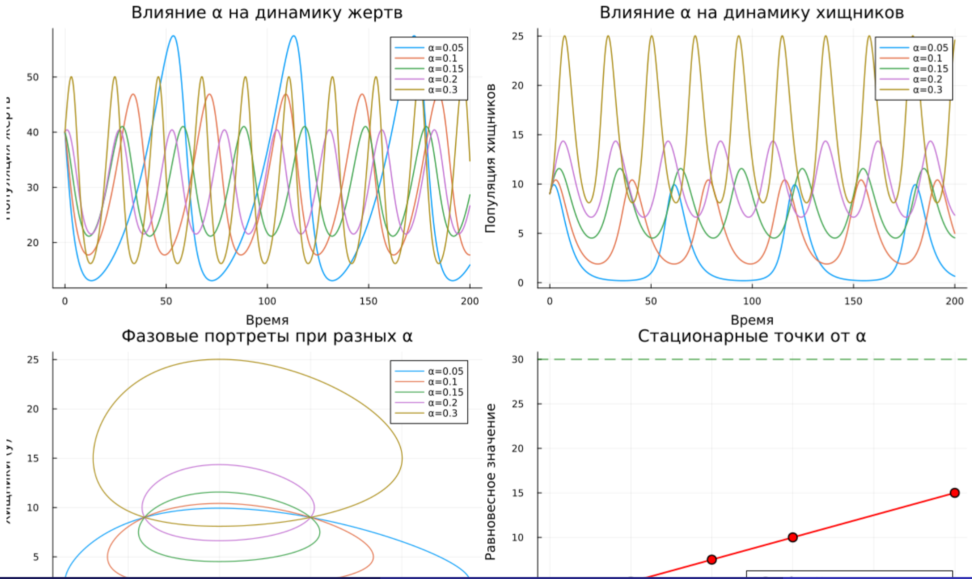
Влияние β на динамику эпидемии
β=0.02, R₀=0.8
β=0.04, R₀=1.6
β=0.06, R₀=2.4
β=0.08, R₀=3.2
β=0.1, R₀=4.0
Время, дни
Зависимость R₀ от β
R_0
β
Пиковое число заражённых от β
Итоговая доля переболевших от β
Доля переболевших, %

: $\alpha \in [0.05, 0.1, 0.15, 0.2, 0.3]$

α	y^*	: –	: –
0.05	2.5	13.1–57.5	0.2–9.9
0.10	5.0	17.8–46.9	1.9–10.4
0.20	10.0	21.5–40.4	6.6–14.4
0.30	15.0	16.2–50.0	8.1–25.0

```
lilidji@eiglushchenko:~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project$ julia --project=. scripts/lv_ode_param.jl
============================================================
Параметрическое исследование модели Лотки-Вольтерры
============================================================
Фиксированные параметры: β = 0.02, δ = 0.01, γ = 0.3
Варьируемый параметр: α ∈ [0.05, 0.1, 0.15, 0.2, 0.3]

α = 0.05: x*=30.0, y*=2.5, жертвы=13.1..57.5, хищники=0.2..9.9
α = 0.1: x*=30.0, y*=5.0, жертвы=17.8..46.9, хищники=1.9..10.4
α = 0.15: x*=30.0, y*=7.5, жертвы=21.2..41.0, хищники=4.5..11.6
α = 0.2: x*=30.0, y*=10.0, жертвы=21.5..40.4, хищники=6.6..14.4
α = 0.3: x*=30.0, y*=15.0, жертвы=16.2..50.0, хищники=8.1..25.0
Could not create decoration from factory! Running with no decorations.

============================================================
Сводная таблица результатов
============================================================
5×9 DataFrame
 Row │ α        x_star   y_star   prey_min  prey_max  prey_mean  pred_min  pred_max  pred_mean
     │ Float64  Float64  Float64  Float64   Float64   Float64    Float64   Float64   Float64
─────┼──────────────────────────────────────────────────────────────────────────────────────────
   1 │    0.05     30.0      2.5   13.0943   57.4634    28.6822  0.202794   9.93079    2.73093
   2 │    0.1      30.0      5.0   17.7539   46.8891    29.7052  1.89517   10.4148     5.20396
   3 │    0.15     30.0      7.5   21.1548   41.0221    29.6608  4.52088   11.5663     7.58309
   4 │    0.2      30.0     10.0   21.5446   40.4222    29.8646  6.63282   14.3501    10.1004
   5 │    0.3      30.0     15.0   16.1898   50.0476    30.5059  8.09488   25.0238    15.0358
```


Фазовые портреты при разных α
Хищники (y)
25
20
15
10
5
α=0.05
α=0.1
α=0.15
α=0.2
α=0.3


Влияние α на динамику жертв
Время
Влияние α на динамику хищников
Популяция хищников
Время
α=0.05
α=0.1
α=0.15
α=0.2
α=0.3
Фазовые портреты при разных α
Стационарные точки от α
Равновесное значение

5

```
lilidji@eiglushchenko:~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project$ julia --project=. scripts/tangle.jl scripts/sir_ode.jl
Генерация из: scripts/sir_ode.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/scripts/sir_ode.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/scripts/sir_ode/sir_ode.jl`
  ✓ Чистый скрипт: /home/lilidji/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/scripts/sir_ode/sir_ode.jl
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/scripts/sir_ode.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/markdown/sir_ode.qmd`
  ✓ Quarto: /home/lilidji/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/markdown/sir_ode.qmd
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/scripts/sir_ode.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/notebooks/sir_ode.ipynb`
  ✓ Notebook: /home/lilidji/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/notebooks/sir_ode.ipynb

Готово! Все файлы созданы.
lilidji@eiglushchenko:~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project$ julia --project=. scripts/tangle.jl scripts/lv_ode.jl
Генерация из: scripts/lv_ode.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/scripts/lv_ode.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/scripts/lv_ode/lv_ode.jl`
  ✓ Чистый скрипт: /home/lilidji/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/scripts/lv_ode/lv_ode.jl
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/scripts/lv_ode.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/markdown/lv_ode.qmd`
  ✓ Quarto: /home/lilidji/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/markdown/lv_ode.qmd
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/scripts/lv_ode.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/notebooks/lv_ode.ipynb`
  ✓ Notebook: /home/lilidji/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/notebooks/lv_ode.ipynb

Готово! Все файлы созданы.
lilidji@eiglushchenko:~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project$ julia --project=. scripts/tangle.jl scripts/sir_ode_param.
Генерация из: scripts/sir_ode_param.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/scripts/sir_ode_param.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/scripts/sir_ode_param/sir_ode_param.jl`
  ✓ Чистый скрипт: /home/lilidji/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/scripts/sir_ode_param/sir_ode_param.jl
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/scripts/sir_ode_param.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/markdown/sir_ode_param.qmd`
  ✓ Quarto: /home/lilidji/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/markdown/sir_ode_param.qmd
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/scripts/sir_ode_param.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/notebooks/sir_ode_param.ipynb`
  ✓ Notebook: /home/lilidji/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project/notebooks/sir_ode_param.ipynb
```

Jupyter Notebooks

```
lilidji@eiglushchenko:~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project$ jupyter nbconvert --to notebook --execute notebooks/lv_ode.ipynb --output lv_ode_e
ed.ipynb
[NbConvertApp] Converting notebook notebooks/lv_ode.ipynb to notebook
Starting kernel event loops.
[NbConvertApp] Writing 3271130 bytes to notebooks/lv_ode_executed.ipynb
lilidji@eiglushchenko:~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project$ jupyter nbconvert --to notebook --execute notebooks/sir_ode_param.ipynb \
--output sir_ode_param_executed.ipynb
[NbConvertApp] Converting notebook notebooks/sir_ode_param.ipynb to notebook
Starting kernel event loops.
[NbConvertApp] Writing 16603 bytes to notebooks/sir_ode_param_executed.ipynb
lilidji@eiglushchenko:~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project$ jupyter nbconvert --to notebook --execute notebooks/sir_ode_param.ipynb \
--output sir_ode_param_executed.ipynb
[NbConvertApp] Converting notebook notebooks/sir_ode_param.ipynb to notebook
Starting kernel event loops.
[NbConvertApp] Writing 16603 bytes to notebooks/sir_ode_param_executed.ipynb
```

4: jupyter nbconvert –execute

```
lilidji@eiglushchenko:~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab02/project$ find . -type f | head -40
./Project.toml
./test/runtests.jl
./src/dummy_src_file.jl
./scripts/sir_ode_param.jl
./scripts/lv_ode_param/lv_ode_param.jl
./scripts/sir_ode/sir_ode.jl
./scripts/lv_ode/lv_ode.jl
./scripts/tangle.jl
./scripts/lv_ode_param.jl
./scripts/sir_ode.jl
./scripts/lv_ode.jl
./scripts/sir_ode_param/sir_ode_param.jl
./plots/sir_panel.png
./plots/lv_ode_param/lv_param_amplitude.png
./plots/lv_ode_param/lv_param_prey.png
./plots/lv_ode_param/lv_param_panel.png
./plots/lv_ode_param/lv_param_phase.png
./plots/lv_ode_param/lv_param_equilibrium.png
./plots/lv_ode_param/lv_param_predator.png
./plots/sir_ode/sir_panel.png
./plots/sir_ode/sir_infected.png
./plots/sir_ode/sir_effective_R.png
./plots/sir_ode/sir_log_scale.png
./plots/sir_ode/sir_phase_portrait.png
./plots/sir_ode/sir_main.png
./plots/sir_ode/sir_percentages.png
./plots/lv_param_amplitude.png
./plots/lv_relative_changes.png
./plots/lv_ode/lv_relative_changes.png
./plots/lv_ode/lv_spectrum.png
./plots/lv_ode/lv_dynamics.png
./plots/lv_ode/lv_derivatives.png
./plots/lv_ode/lv_phase_portrait.png
./plots/lv_ode/lv_panel.png
./plots/lv_param_prey.png
./plots/lv_param_panel.png
```

6

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6.